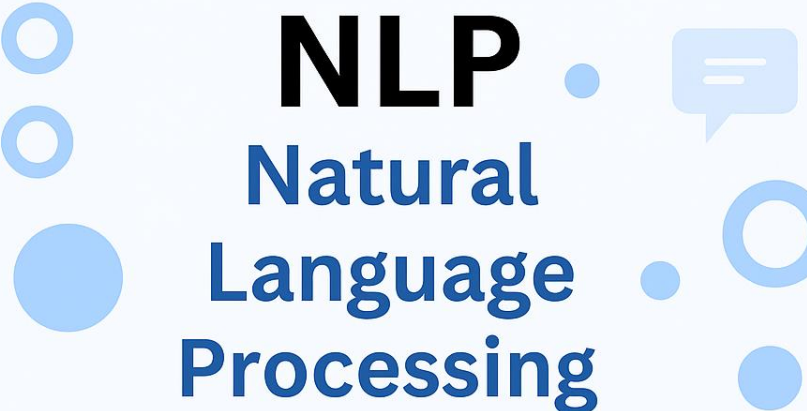
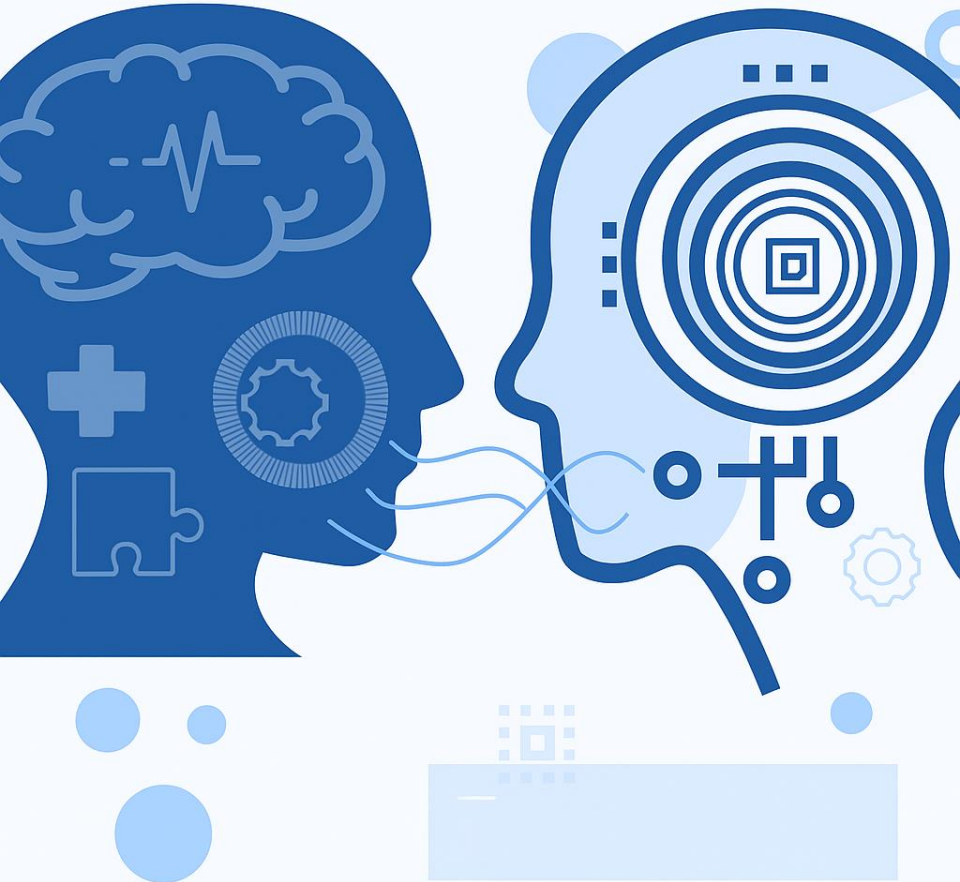




NLP

Natural
Language
Processing



NATURAL LANGUAGE PROCESSING (NLP)

PMDS606L

MODULE 4

LECTURE 2

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CLOSED CLASSES

- **Prepositions:** on, under, over, near, by, at, from, to, with
- **Particles:** up, down, on, off, in, out, at, by
- **Determiners:** a, an, the
- **Conjunctions:** and, but, or, as, if, when
- **Pronouns:** she, who, I, others
- **Auxiliary verbs:** can, may, should, are
- **Numerals:** one, two, three, first, second, third

PREPOSITIONS

Prepositions occur before noun phrases. Semantically they often indicate spatial or temporal relations, whether literal (***on it, before then, by the house***) or metaphorical (***on time, with gusto, beside herself***), but often indicate other relations as well, like marking the agent in ***Hamlet was written by Shakespeare.***

A **particle** resembles a preposition or an adverb and is used in combination with a verb. Particles often have extended meanings that aren't quite the same as the prepositions they resemble, as in the particle ***over*** in ***she turned the paper over.***



PHRASAL VERB

A verb and a particle that act as a single syntactic and/or semantic unit are called a **phrasal verb**. The meaning of phrasal verbs is often problematically non-compositional – not predictable from the distinct meanings of the verb and the particle.

- ***turn down*** means '***reject***'
- ***rule out*** means '***eliminate***'
- ***find out*** means '***discover***'
- ***go on*** means '***continue***'

DETERMINER

A closed class that occurs with nouns, often marking the beginning of a noun phrase, is the **determiner**. One small subtype of determiners is the **article**: English article has three articles: *a*, *an*, and *the*. Other determiners include *this* and *that* (*this chapter*, *that page*). *A* and *an* mark a noun phrase as indefinite, while *the* can mark it as definite. Articles are quite frequent in English; indeed, *the* is the most frequently occurring word in most corpora of written English, and *a* and *an* are generally right behind.

CONJUNCTIONS

Conjunctions join two phrases, clauses, or sentences. **Coordinating conjunctions** like *and*, *or*, and *but* join two elements of equal status. **Subordinating conjunctions** are used when one of the elements has some embedded status. For example, *that* in “*I thought that you might like some milk*” is a **subordinating conjunction** that links the main clause *I thought* with the subordinate clause *you might like some milk*. This clause is called subordinate because this entire clause is the “content” of the main verb thought. Subordinating conjunctions like *that* which link a verb to its argument in this way are also called **complementizers**.



PRONOUNS

Pronouns are forms that often act as a kind of shorthand for referring to some noun phrase or entity or event. **Personal pronouns** refer to persons or entities (*you, she, I, it, me*, etc.). **Possessive pronouns** are forms of personal pronouns that indicate either actual possession or more often just an abstract relation between the person and some object (*my, your, his, her, its, one's, our, their*). **Wh-pronouns** (*what, who, whom, whoever*) are used in certain question forms, or may also act as **complementizers** (*Frida, who married Diego. . .*).

AUXILIARY VERBS

- A closed class subtype of English verbs is the **auxiliary verbs**.
- Cross-linguistically, auxiliaries mark semantic features of a main verb: whether an action takes place in the present, past, or future (**tense**), whether it is completed (**aspect**), whether it is negated (**polarity**), and whether an action is necessary, possible, suggested, or desired (**mood**).
- English auxiliaries include the **copula** verb **be**, the two verbs **do** and **have**, along with their inflected forms, as well as a class of **modal** verbs.
- **Be** is called a **copula** because it connects subjects with certain kinds of predicate nominals and adjectives (**He is a duck**). The verb **have** can mark the perfect tenses (**I have gone, I had gone**), and **be** is used as part of the passive (**We were robbed**) or progressive (**We are leaving**) constructions.
- **Modals** are used to mark the mood associated with the event depicted by the main verb: **can** indicates ability or possibility, **may** indicates permission or possibility, **must** indicates necessity. There is also a modal use of **have** (e.g., **I have to go**).

OTHER POS

English also has many words of more or less unique function, including **interjections** (*oh, hey, alas, uh, um*), **negatives** (*no, not*), **politeness markers** (*please, thank you*), **greetings** (*hello, goodbye*), and the **existential there** (*there are two on the table*), among others. These classes may be distinguished or lumped together as interjections or adverbs, depending on the purpose of the labelling.

TAGSETS

- More recent lists of parts-of-speech (or tagsets) have many more word classes –
- 45 for the **Penn Treebank** (Marcus et al., 1993)
- 87 for the **Brown corpus** (Francis, 1979; Francis and Kučera, 1982)
- 146 for the **C7 tagset** (Garside et al., 1997).

PENN TREEBANK

Tag	Description	Example	Tag	Description	Example	Tag	Description	Example
CC	coordinating conjunction	<i>and, but, or</i>	PDT	predeterminer	<i>all, both</i>	VBP	verb non-3sg present	<i>eat</i>
CD	cardinal number	<i>one, two</i>	POS	possessive ending	<i>'s</i>	VBZ	verb 3sg pres	<i>eats</i>
DT	determiner	<i>a, the</i>	PRP	personal pronoun	<i>I, you, he</i>	WDT	wh-determ.	<i>which, that</i>
EX	existential 'there'	<i>there</i>	PRP\$	possess. pronoun	<i>your, one's</i>	WP	wh-pronoun	<i>what, who</i>
FW	foreign word	<i>mea culpa</i>	RB	adverb	<i>quickly</i>	WP\$	wh-possess.	<i>whose</i>
IN	preposition/ subordin-conj	<i>of, in, by</i>	RBR	comparative adverb	<i>faster</i>	WRB	wh-adverb	<i>how, where</i>
JJ	adjective	<i>yellow</i>	RBS	superlatv. adverb	<i>fastest</i>	\$	dollar sign	<i>\$</i>
JJR	comparative adj	<i>bigger</i>	RP	particle	<i>up, off</i>	#	pound sign	<i>#</i>
JJS	superlative adj	<i>wildest</i>	SYM	symbol	<i>+, %, &</i>	"	left quote	<i>' or "</i>
LS	list item marker	<i>1, 2, One</i>	TO	"to"	<i>to</i>	"	right quote	<i>' or "</i>
MD	modal	<i>can, should</i>	UH	interjection	<i>ah, oops</i>	(	left paren	<i>[, (, {, <</i>
NN	sing or mass noun	<i>llama</i>	VB	verb base form	<i>eat</i>	)	right paren	<i>],), }, ></i>
NNS	noun, plural	<i>llamas</i>	VBD	verb past tense	<i>ate</i>	,	comma	<i>,</i>
NNP	proper noun, sing.	<i>IBM</i>	VBG	verb gerund	<i>eating</i>	.	sent-end punc	<i>. ! ?</i>
NNPS	proper noun, plu.	<i>Carolinas</i>	VBN	verb past part.	<i>eaten</i>	:	sent-mid punc	<i>: ; ... - -</i>

POS TAGGING

The grand jury commented on a number of other topics.

Determiners - the and a

Adjectives - grand and other

Common nouns - jury, number, and topics

Past tense verb - commented.

**The/DT grand/JJ jury/NN commented/VBD on/IN a/DT
number/NN of/IN other/JJ topics/NNS ./.**

POS TAGGING

There are 70 children there

EX - existential there

Adverb(RB) - there

There/EX are/VBP 70/CD children/NNS there/RB

POS TAGGING

Preliminary findings were reported in today's New England Journal of Medicine.

Segmentation of the possessive morpheme 's.

A passive construction, '***were reported***', in which ***reported*** is tagged as a past participle(VBN).

New England Journal of Medicine is a proper noun; the Treebank tagging chooses to mark each noun in it separately as NNP, including ***journal*** and ***medicine***, which might otherwise be labelled as common nouns (NN).

**Preliminary/JJ findings/NNS were/VBD reported/VBN in/IN
today/NN's/POS New/NNP England/NNP Journal/NNP of/IN
Medicine/NNP ./.**

POS TAGGING

- Tagging is a disambiguation task.
- Words are ambiguous – have more than one possible part-of-speech – and the goal is to find the correct tag for the situation.
- Book can be a verb (**book that flight**) or a noun (**hand me that book**).
- That can be a determiner (**Does that flight serve dinner**) or a complementizer (**I thought that your flight was earlier**).
- The goal of POS-tagging is to resolve these ambiguities, choosing the proper tag for the context.

AMBIGUITY

- How common is tag ambiguity?
- Most word types (85-86%) are unambiguous (Janet is always NNP, funniest JJS, and hesitantly RB).
- But the ambiguous words, though accounting for only 14-15% of the vocabulary, are very common words, and hence 55-67% of word tokens in running text are ambiguous.

Types:		WSJ		Brown	
Unambiguous	(1 tag)	44,432	(86%)	45,799	(85%)
Ambiguous	(2+ tags)	7,025	(14%)	8,050	(15%)
Tokens:					
Unambiguous	(1 tag)	577,421	(45%)	384,349	(33%)
Ambiguous	(2+ tags)	711,780	(55%)	786,646	(67%)

AMBIGUITY

Some of the most ambiguous frequent words are that, back, down, put and set; here are some examples of the 6 different parts of speech for the word back:

- earnings growth took a **back/JJ** seat
- a small building in the **back/NN**
- a clear majority of senators **back/VBP** the bill
- Dave began to **back/VB** toward the door
- enable the country to buy **back/RP** about debt
- I was twenty-one **back/RB** then

AMBIGUITY RESOLUTION

- **Most Frequent Class Baseline:** Always compare a classifier against a baseline at least as good as the most frequent class baseline (assigning each token to the class it occurred in most often in the training set).
- How good is this baseline? A standard way to measure the performance of part-of-speech taggers is accuracy: the percentage of tags correctly labelled (matching human labels on a test set). If we train on the WSJ training corpus and test on sections 22-24 of the same corpus, the most-frequent-tag baseline achieves an accuracy of 92.34%. By contrast, the state of the art in part-of-speech tagging on this dataset is around 97% tag accuracy, a performance that is achievable by most algorithms (HMMs, MEMMs, neural networks, rule-based algorithms).

MARKOV PROPERTY

In probability theory and statistics, the term Markov property refers to the memoryless property of a stochastic process, which means that its future evolution is independent of its history; it only depends on the current state. It is named after the Russian mathematician Andrey Markov.

MARKOV MODEL

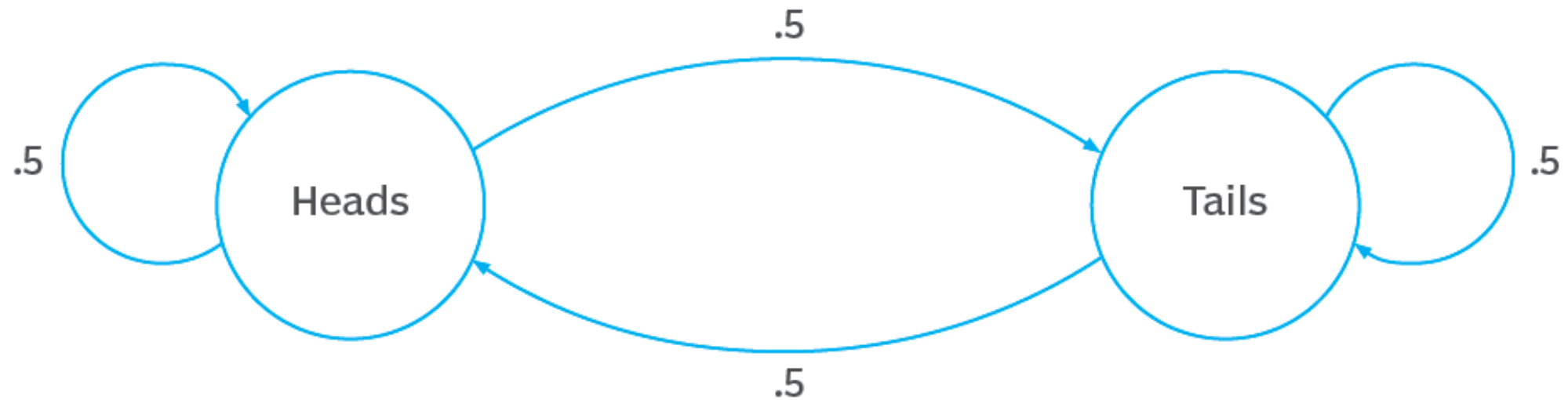
A model that follows the Markov property.

Markov models

	System state is fully observable	System state is partially observable
System is autonomous	Markov chain	Hidden Markov model
System is controlled	Markov decision process	Partially observable Markov decision process

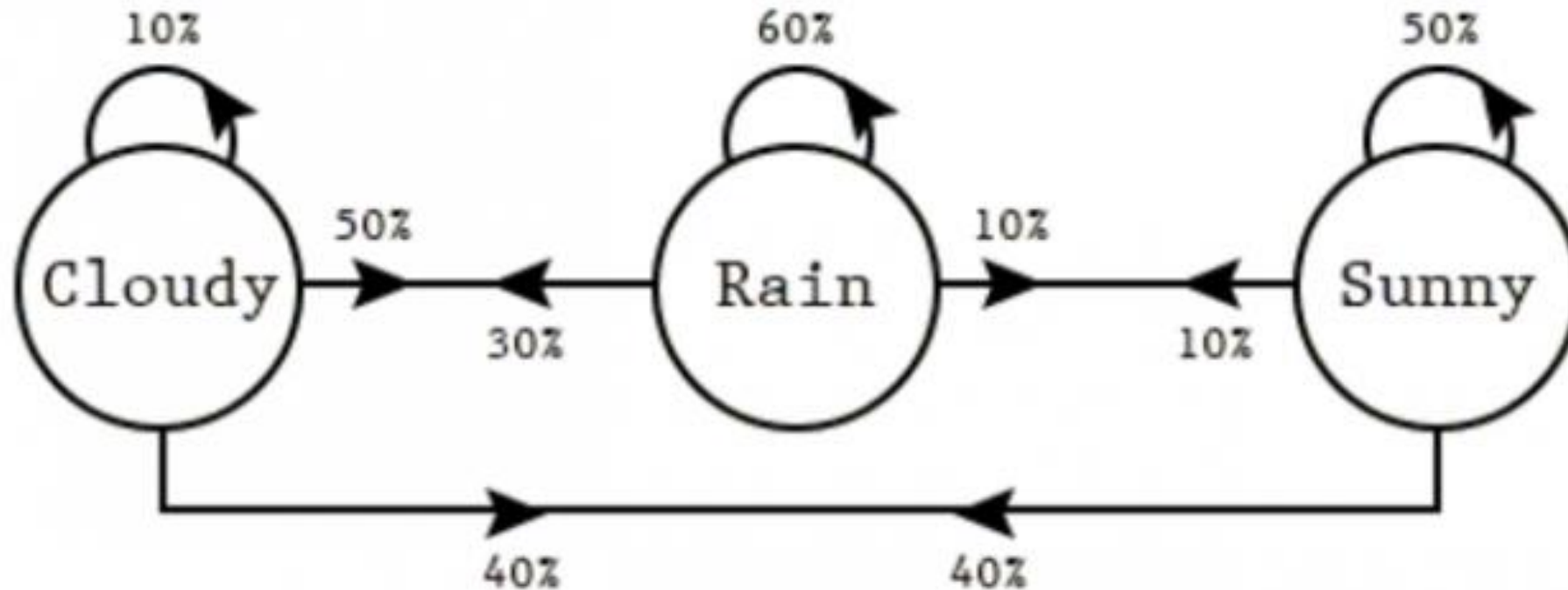
MARKOV CHAIN EXAMPLE

Markov chain coin toss



MARKOV CHAIN EXAMPLE

What is the probability that Wednesday will be cloudy if Monday is sunny?



HIDDEN MARKOV MODEL (HMM)

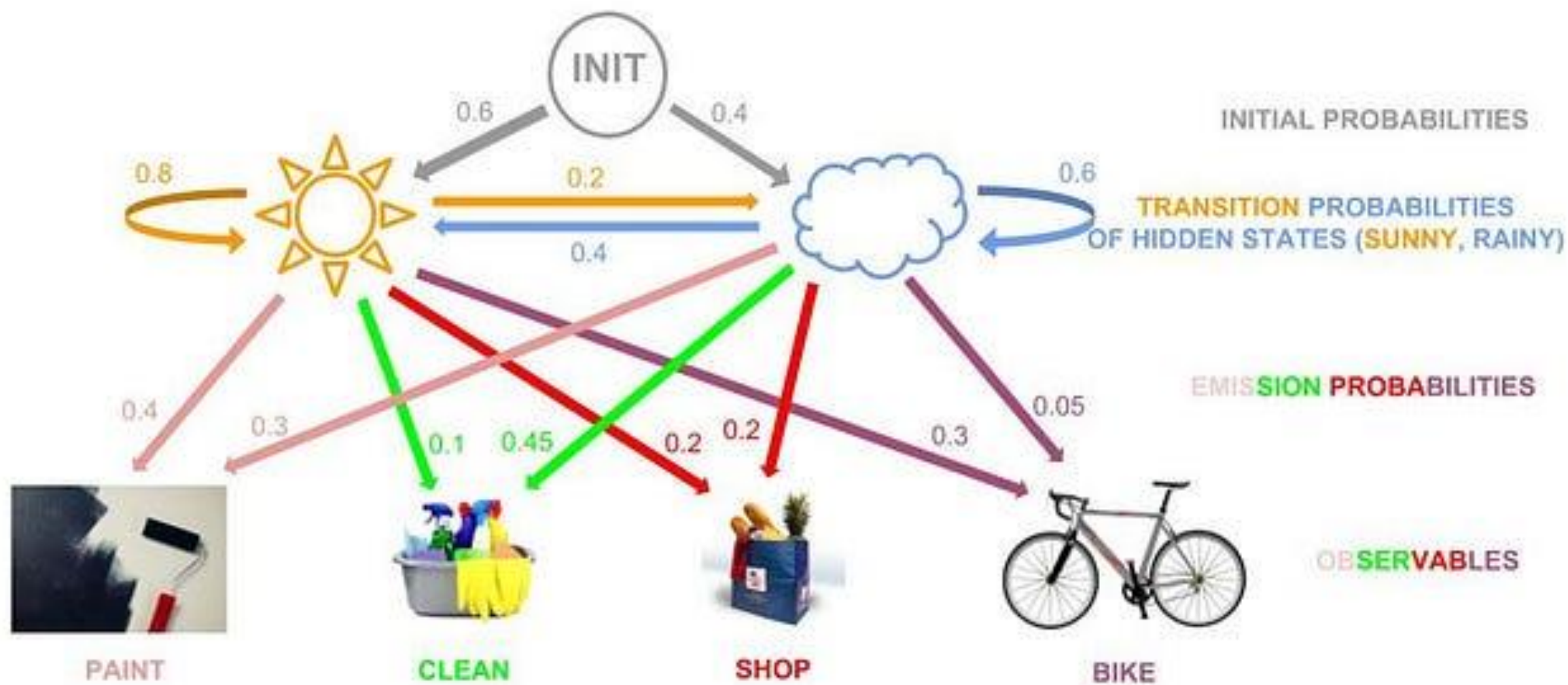
Markov chain with unobservable/hidden states.

Lea is our imaginary friend and during the day she does either of these four things:

- Painting
- Cleaning the house
- Biking
- Shopping for groceries

} **Observable States**

From Lea's activities, we want to know whether the day has been sunny or rainy. These two are going to be our **hidden states**.



TRANSITION PROBABILITIES

$A =$

		to	
from			
		0.8	0.2
		0.4	0.6

EMISSION PROBABILITIES

$B =$

	from	to			
		PAINT 	CLEAN 	SHOP 	BIKE 
		0.4	0.1	0.2	0.3
		0.3	0.45	0.2	0.05

HMM ELEMENTS

$$S = \{ S_{\text{sunny}}, S_{\text{rainy}} \} \text{ (Hidden States)}$$

$$O = \{ O_{\text{clean}}, O_{\text{bike}}, O_{\text{shop}}, O_{\text{paint}} \} \text{ (Observables)}$$

$$\pi = [0.6 \ 0.4] \text{ (Initial Probabilities)}$$

$$A = \begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix} \text{ (Transition Probabilities)}$$

$$B = \begin{bmatrix} 0.4 & 0.1 & 0.2 & 0.3 \\ 0.3 & 0.45 & 0.2 & 0.05 \end{bmatrix} \text{ (Emission Probabilities)}$$

BUILD HMM

```
states = ('Rainy', 'Sunny')

observations = ('walk', 'shop', 'clean')

start_probability = {'Rainy': 0.6, 'Sunny': 0.4}

transition_probability = {
    'Rainy' : {'Rainy': 0.7, 'Sunny': 0.3},
    'Sunny' : {'Rainy': 0.4, 'Sunny': 0.6},
}

emission_probability = {
    'Rainy' : {'walk': 0.1, 'shop': 0.4, 'clean': 0.5},
    'Sunny' : {'walk': 0.6, 'shop': 0.3, 'clean': 0.1},
}
```

